ATROPOS CORE ENGINE GAP MAP v2

Authority: Source Docs 1–3 + 100% Completion Blueprint (Phases 0–19) + 29 Jul 2026 export (file_count=1475) Density: raw atoms · fingerprints · predicates · edges · RESEARCH · IMPL HOE / Phase 20 full learning: separate documents Non-source superiority primitives: marked [NON-SOURCE]

0. AUTHORITY LEDGER

- Blueprint hash: (export-derived)
- Source Doc 1–3: present in attachments
- Export generated_utc=2026-07-29T13:24:57Z · file_count=1475
- Critical closed: ConstraintSolverEvaluator=36L · TokenIsolationVault=149L · TreeSitterGrammarBridge=83L · ScaffoldAdapters=3L (split)
- SelfHostInsideOutSandboxProofTest=261L present
- AgentCommand=406L · DloiService=345L (atomic-decoupling targets)

1. CHECKPOINT 1 — SELF-BUILD 100% + PHASES 0–11 (no stubs) Acceptance: From running JAR: NL prompt for change → provider communication → verified patch → compile gate → real shell/bash mutation of ATROPOS source → git status proof → optional add/commit/push → optional rebuild prompt. Existing sandbox proof counts. Phone-side full rebuild not required.

C1-SB-01 | Phase11+SelfHost | NL cradle prompt inside JAR mutates real source via shell bounds | SelfHostInsideOutSandboxProofTest 261L + SelfHost* suite | PARTIAL | C1-SB-02 RESEARCH: Trace SelfHostInsideOutSandboxProofTest to the exact interactive JAR entrypoint that must accept NL and call shell-bounded mutation; list remaining stubs between test and live path. IMPL: Wire existing SelfHostCommand + worktree bounds; compile gate before any write; proof via git status inside JAR.

C1-SB-02 | Phase11 | Compile gate before mutation; nonzero exit forbids VERIFIED | VerifiedCompletionGate + tests | PARTIAL | C1-SB-03 RESEARCH: Confirm every self-mutation path already calls VerifiedCompletionGate; identify any bypass. IMPL: Nonzero compile/test exit must block VERIFIED and promotion; reuse existing gate, no second verifier.

C1-SB-03 | Phase11 | git status + optional commit/push of self-changes from inside JAR | SelfHostPromotionService present | PARTIAL | — RESEARCH: Locate SelfHostPromotionService call sites; define minimal git status/commit/push proof that runs inside JAR without full phone rebuild. IMPL: After verified mutation, surface git status; optional add/commit/push through existing shell bridge; rebuild prompt only, not forced.

C1-P0 | Phase0 | Fresh clone Termux + CI → identical accepted JAR | PARTIAL/VERIFY | PARTIAL | C1-P1 RESEARCH: Pin exact Kotlin/JDK/Gradle matrix and fingerprints; list any tracked residue that breaks reproducibility. IMPL: Clean fingerprints into docs; CI and Termux must produce same accepted JAR hash.

C1-P1 | Phase1 | Multi-field doctor report; never ready on descriptor/key alone | STRONG PARTIAL | PARTIAL | C1-P2 RESEARCH: Diff ProviderActivationService fields against required readiness matrix (descriptor/adapter/fixture/dry-run/key/live/quota/route). IMPL: Unify into one durable doctor report; ready only when all fields proven.

C1-P2 | Phase2 | Normalized transports; ScaffoldAdapters fully split; fixtures pass | PARTIAL (ScaffoldAdapters=3L split) | PARTIAL | C1-P3 RESEARCH: Confirm remaining transport/normalization/error-mapping files; list any still-mixed adapter. IMPL: Finish atomic split; every outcome fixture (success/auth/rate/billing/timeout/malformed/empty/cancel/redaction) must pass offline.

C1-P3 | Phase3 | Route law enforced; free-first; explainable skips; no accidental paid | STRONG PARTIAL | PARTIAL | C1-P4 RESEARCH: Trace RoutePolicy + FreeModeGuard + EmergencyPaidGate; prove paid path is opt-in only. IMPL: Record every usage/failure; queue on free exhaustion; route explanation shows selected and skipped.

C1-P4 | Phase4 | TokenIsolationVault real; secrets never appear in any persisted/displayed surface | TokenIsolationVault=149L | PARTIAL | C1-P5 RESEARCH: Confirm vault isolation and RedactionFilter call sites before UI/logs/queue/memory/evidence. IMPL: Real local isolation; any confirmed leak is release-blocking.

C1-P5 | Phase5 | Offline fixture matrix complete for every registered adapter | PARTIAL/VERIFY | PARTIAL | C1-P6 RESEARCH: List adapters missing success/auth/rate/redaction/dry-run fixtures. IMPL: Offline matrix is source of truth; live tests opt-in only.

C1-P6 | Phase6 | DLOI exact authority; typed NoMatch; no blind RAG | STRONG PARTIAL (DloiService=345L) | PARTIAL | C1-P7 RESEARCH: Confirm HigZeroGuard callers; list any remaining ad-hoc exception paths that discard reason. IMPL: Every requirement resolves to exact section or typed NoMatch; search never becomes authority.

C1-P7 | Phase7 | Real deterministic parsing feeds AstSymbolGraph; symbol census reproducible | TreeSitterGrammarBridge=83L | PARTIAL | C1-P8 RESEARCH: Verify parser offsets/packages/classes/functions/imports/callers now populate graph; list missing impact queries. IMPL: Deterministic parse only; no probabilistic symbol recovery.

C1-P8 | Phase8 | ConstraintSolverEvaluator real; IndependentVerificationGate facade; no self-approval | ConstraintSolverEvaluator=36L | PARTIAL | C1-P9 RESEARCH: Confirm constant-true paths removed; map IndependentVerificationGate over existing DeterministicVerifier + VerifiedCompletionGate. IMPL: Failed assertion blocks completion; each finding names invariant/expected/observed/evidence.

C1-P9 | Phase9 | CAS memory; failure signatures; never overrides authority | PARTIAL | PARTIAL | C1-P10 RESEARCH: Locate LocalMemoryStore CAS/hash paths; confirm secret exclusion and authority non-override. IMPL: Memory survives restart; queryable by exact coordinates; never becomes law.

C1-P10 | Phase10 | ActionProposal → BoundedAgencyGate → TypedToolExecutor; no raw prose execution | PARTIAL | PARTIAL | C1-SB-01 RESEARCH: Inventory current policy/tool paths; identify any remaining raw shell/prose execution. IMPL: Every side effect: schema → context → source/DAG → territory → redaction → policy → typed tool → verifier → evidence.

C1-X1 | continuous | Extreme per-file atomic decoupling; ArchitectureComplianceChecker blocking | AgentCommand=406L · DloiService=345L | OPEN | all C1 RESEARCH: Run line+concern scan; list remaining mixed-concern files above threshold. IMPL: One file = one responsibility; violations advisory then blocking; every batch leaves tree better or equal.

2. CHECKPOINT 2 — APP FACTORY 100% + PHASES 12–16 Acceptance: Natural-language calculator → real code → GitHub new repo (tests/README/LICENSE/.gitignore/AGENTS.md complete) + Phases 12–16 complete.

C2-P12 | Phase12 | Director advisory; scope drift detected before promotion | PARTIAL | PARTIAL | C2-P13 RESEARCH: Bind Director observations to goals/claims/worktrees/territories; confirm advisory never generates executable facts. IMPL: Low-cost diff monitoring; auditable advisory only.

C2-P13 | Phase13 | TerritoryEnforcer blocks out-of-territory mutation before it happens | PARTIAL | PARTIAL | C2-P14 RESEARCH: Trace TerritoryService + IsolatedWorktreeService; confirm pre-mutation checks on create/edit/delete/rename/shell. IMPL: Territory recorded at dispatch; violations blocking in enforcement mode.

C2-P14 | Phase14 | HR Router sole cross-boundary channel; redacted, audited | PARTIAL | PARTIAL | C2-P15 RESEARCH: Confirm no free-form agent mailboxes remain as coordination layer. IMPL: All cross-territory requests through HR; approved/narrowed/redacted/denied/escalated with audit.

C2-P15 | Phase15 | Auditor blocks promotion independently; Custodian safe cleanup only | PARTIAL | PARTIAL | C2-P16 RESEARCH: Bind Auditor findings to completion gates; list Custodian allowed cleanup policies. IMPL: Neither may silently mutate authority or approve own work.

C2-P16 | Phase16 | Manager/Specialist/Worker hierarchy; dispatch carries territory/capabilities/budget/acceptance/rollback | EARLY PARTIAL | OPEN | C3-AF-01 RESEARCH: Retire or rework DirectorOrchestrator/WorkerCodeSynthesizer paths that invent file plans or write directly. IMPL: Director decomposes; Manager dispatches; bounded Specialist/Worker executes; Auditor verifies; Human final authority.

3. CHECKPOINT 3 — PHASES 17–18 + APP FACTORY + PHASE 19 Acceptance: CLI + local-server Web + Android APK functioning for Phases 17–18; App Factory natural-language full-stack with live proof; Phase 19 complete.

C3-P17 | Phase17 | Multimodal: isolated preview, screenshots, visual comparison, accessibility checks | EARLY PARTIAL | OPEN | C3-P18 RESEARCH: Locate existing snapshot/inspection paths; define controlled browser actuator bounds. IMPL: Launch isolated preview; capture evidence; attach to originating requirement; no unrestricted host access.

C3-P18 | Phase18 | Multiplatform clients (CLI/Web/Desktop/Android) share core; no client-specific orchestration fork | MISSING/VERY EARLY | OPEN | C3-AF-01 RESEARCH: Inventory KMP/shared-core boundaries already present; list duplicated logic. IMPL: Presentation only on clients; same durable project/run controllable from all surfaces.

C3-AF-01 | Phase19 | NL project creation → real source tree + Git history | EARLY PARTIAL | OPEN | C3-AF-02 RESEARCH: Map AppFactoryRouter + artifact pipeline to greenfield project creation path. IMPL: One NL request creates project with real paths and Git history; no plan-only success.

C3-AF-02 | Phase19 | Live preview + hot reload + diagnostics + rollback | EARLY PARTIAL | OPEN | C3-AF-03 RESEARCH: Compose LivePreviewService/BrowserActuator over existing snapshot/inspection. IMPL: Preview is acceptance requirement; blank/error states visible; rollback preserved.

C3-AF-03 | Phase19 | Full-stack generation (frontend/backend/auth/db/storage) with security rules | EARLY PARTIAL | OPEN | C3-AF-04 RESEARCH: List missing atomic components (ProjectRegistry/RepositoryBinding/GitHubBinding/BoundedWorkExecutor/EvidenceCollector/BatchGate). IMPL: Compose over existing DAG/territory/verifier; never duplicate.

C3-AF-04 | Phase19 | Browser-driven user-flow tests + deterministic backend verification | EARLY PARTIAL | OPEN | C3-AF-05 RESEARCH: Define acceptance clips for real user flows; bind to EvidenceCollector. IMPL: Tests and verification required before commit/promote.

C3-AF-05 | Phase19 | GitHub new repo + tests/README/LICENSE/.gitignore/AGENTS.md complete; portable ownership | EARLY PARTIAL | OPEN | C3-P19 RESEARCH: Define complete-repo predicate checklist; map to existing GitHub binding paths. IMPL: One NL prompt yields complete repo; user owns it; ATROPOS does not retain required cloud dependency.

C3-P19 | Phase19 | Activity monitor shows every plan/provider/tool/diff/test/verifier/artifact/deploy state | EARLY PARTIAL | OPEN | C4-IF-01 RESEARCH: Unify journal + verification + artifact + run-observer into single observable stream. IMPL: Monitor is presentation of existing evidence; no second event system.

4. CHECKPOINT 4 — PHASE 20 INTERFACE HOOKS (full Phase 20 architecture is separate document) C4-IF-01 | Phase20 interface | Evidence ledger presentation for runtime observations | PARTIAL | OPEN | — RESEARCH: Define evidence → memory → proposal schema already partially present; surface only. IMPL: Observations become memory; never automatic law.

C4-IF-02 | Phase20 interface | Proposal gate UI; accepted authority only after verifier/auditor | PARTIAL | OPEN | — RESEARCH: Map Auditor gate to proposal acceptance flow. IMPL: Proposals require independent acceptance; original source docs remain immutable unless explicitly superseded.

C4-IF-03 | Phase20 interface | Versioned amendment hash display; re-verification on authority change | PARTIAL | OPEN | — RESEARCH: Locate version/hash fields for amendments. IMPL: Accepted authority gets own versioned hash; re-verify dependents.

C4-IF-04 | Phase20 interface | Reproducibility predicate + before/after metric declaration | PARTIAL | OPEN | — RESEARCH: Define frequency/threshold for deficiency to qualify as proposal; require metric target before code. IMPL: Transient hiccups never become law; improvement must declare metric before implementation.

C4-IF-05 | Phase20 interface | Anti-oscillation cooldown + hard boundary display | PARTIAL | OPEN | — RESEARCH: Existing cooldown patterns in route/quota; extend to self-improvement proposals. IMPL: Cooldowns and hard boundaries on governance invariants; never silent self-modification of authority.

5. CONTINUOUS + [NON-SOURCE] CONT-01 | continuous | ArchitectureComplianceChecker expanded; line+concern thresholds | OPEN RESEARCH: Implement configurable thresholds; scan for routing+rendering, transport+normalization, verification+execution mixes. IMPL: Advisory then blocking; every batch improves or holds atomicity.

CONT-02 | continuous | Source-to-code ledger + symbol census updated every batch | OPEN RESEARCH: Automate ledger update from DLOI + AstSymbolGraph. IMPL: No batch closes without ledger and census refresh.

NS-01 | [NON-SOURCE] | Cryptographically Verifiable Agent Authorization (CVA) + proof-carrying bundles | OPEN RESEARCH: Ed25519 + Merkle patterns for capability tokens and evidence bundles; compare to existing attestation. IMPL: Capability tokens minted with proof; verification collapses to hash under determinism.

NS-02 | [NON-SOURCE] | Proof-carrying execution bundles for every promotion and self-change | OPEN RESEARCH: Bundle schema (patch + tests + verifier findings + hashes) already partially present. IMPL: Promotion requires proof-carrying bundle; no naked success.

NS-03 | [NON-SOURCE] | Determinism Thesis: prefer platform-deterministic paths; verification O(1) hash | OPEN RESEARCH: Identify remaining non-deterministic inference paths in critical gates. IMPL: Prefer bitwise-identical execution where feasible; collapse verification cost.

NS-04 | [NON-SOURCE] | Harness-as-Asset + Unified Assertion Interface | OPEN RESEARCH: Domain invariants as machine-readable registries; closed-loop monotonic convergence. IMPL: Assertions are data; harness improves without changing law.

NS-05 | [NON-SOURCE] | Provable algebraic deadlock on unsafe transitions | OPEN RESEARCH: Model unsafe state reachability; induction that unsafe states unreachable under gates. IMPL: Territory + policy + verifier form algebraic barrier; document unreachable proofs.

NS-06 | [NON-SOURCE] | Canonical-code / No-Accident Horizon constraint | OPEN RESEARCH: Reduce frontier-model inventiveness surface; constrain agents to verified machine. IMPL: Agents operate inside constrained software machine; inventiveness outside territory is refused.

6. ORDERED GAP DAG C1-P0 → C1-P1 → C1-P2 → C1-P3 → C1-P4 → C1-P5 → C1-P6 → C1-P7 → C1-P8 → C1-P9 → C1-P10 → C1-SB-01 → C1-SB-02 → C1-SB-03 C1-X1 blocks all C1 C1-SB-03 → C2-P12 → C2-P13 → C2-P14 → C2-P15 → C2-P16 C2-P16 → C3-P17 → C3-P18 → C3-AF-01 → C3-AF-02 → C3-AF-03 → C3-AF-04 → C3-AF-05 → C3-P19 C3-P19 → C4-IF-01..05 CONT-01, CONT-02, NS-01..06 continuous across all checkpoints

7. RAW CLOSURE METRICS Checkpoint 1 critical stubs: CLOSED Checkpoint 1 self-build mechanical path: PARTIAL (sandbox proof exists; interactive JAR mutation + git push still open) Checkpoint 2: all PARTIAL / EARLY Checkpoint 3: EARLY / OPEN Checkpoint 4 interface: PARTIAL Continuous atomic decoupling: OPEN (AgentCommand 406L, DloiService 345L) [NON-SOURCE] superiority: all OPEN

Every atom above carries RESEARCH (one-line LLM prompt) and IMPL (1–2 line necessary note). Pursue RESEARCH before coding when gap is OPEN.

End of core gap map v2. Phase 20 full rule set and HOE are separate documents.